LUẬN VĂN TỐT NGHIỆP

CO4337 | HK252

Ứng dụng DevOps và container hóa trên Kubernetes cho hệ thống Moodle LMS tích hợp AI

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Overview & problem identification

Overview

- **Context:** Digital transformation trends in education
- **Need:** A flexible and highly scalable LMS system
- **Solution:** Moodle on Kubernetes with integrated AI
- **Technologies:** Cloud-native, containerization, and automation

Objectives

1. Deploy Moodle LMS on Kubernetes

- High concurrency handling
- Auto-scaling capability
- High availability & fault tolerance
- **Scope:** Multi-cloud

2. Build a complete CI/CD pipeline

- Automated build, test, and deployment

3. Integrate an AI subsystem

- Local AI (Ollama) or Cloud AI (OpenAI, Gemini, etc.) to assist learning

Moodle LMS

Open-source learning management system

Pros

- Highly flexible and customizable
- Large, active development community
- Extensive support for various plugins and themes

Target Version: Moodle 5.x

Tech Stack: PHP 8.2-FPM, PostgreSQL 15, Redis

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Analysis, Design & Implementation

Container orchestration platform

Comparison table

Criteria	Cloud Managed Services	Kubernetes
Vendor Lock-in	**High** Cloud-dependent, Costly to migrate	**None** Cloud-agnostic, On-prem ready
Operational Overhead	**Minimal** Managed by provider	**High** Requires skilled engineers
Customizability	**Limited** by provider	**Unlimited** Highly extensible
Time-to-Market	**Rapid** Ideal for immediate deployment	**Slow** initial setup Requires upfront design

Infrastructure cost analysis

General configuration

- Autoscale 3-6 Node

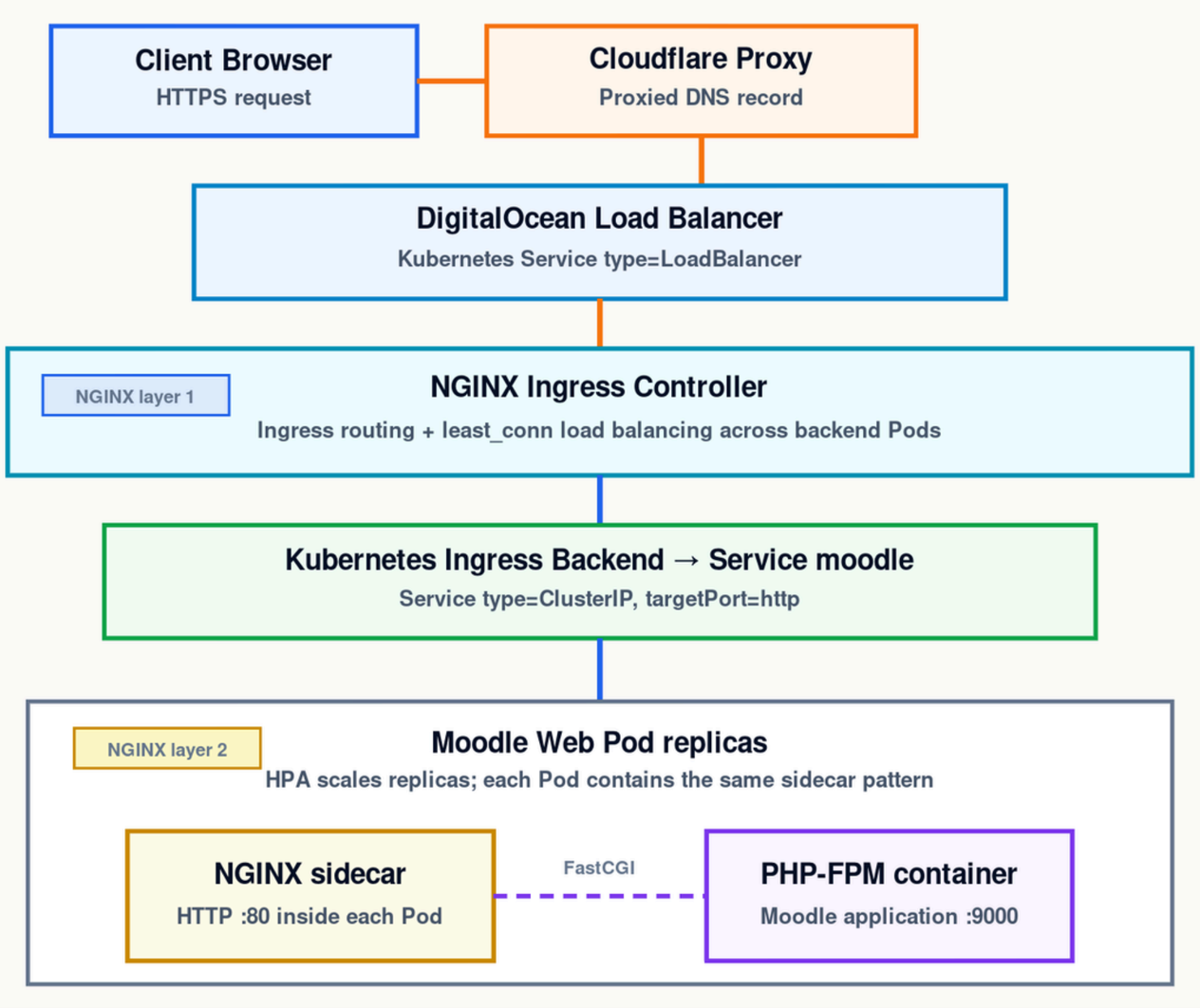
(Each node 4VCPu, 8GB RAM)

- 2 Databases designed using Primary + Standby.

(Each DB 2 VCPu, 4GB RAM, max 97 connection pools))

(Cost/month)	DigitalOcean (DO)	AWS	Azure
K8s Basic	0$	73$	0$
Node Scale 3→6	48$ 144-288	~124$ 372-744	~124$ 370-740
Database	$122	~$150	~$140
Load balancer	$15.00	~$30.00	~$20.00
Tổng	**~281 - 425$**	**~625 - 997$**	**~530 - 900$**

Web Server Architecture Comparison



Comparison table of Nginx vs. Apache

Criteria	Nginx Reverse Proxy & PHP-FPM	Apache Web Server (Monolithic)
I/O Mechanism	**Event-driven** Non-blocking	**Process-based** Blocking
Connection Management	**1** worker **multiple** connections	**1** process **1** connection
Processing	**Decoupled** (Static/Dynamic split)	**Mixed** (One process handles all)
Resource Use	**Optimized** (Stable RAM)	**High** (Linear RAM growth)
Scalability	**Easy** Independent scaling	**Difficult** High bottleneck risk

Package Manager

Helm is the standard package manager for k8s, helping to package all the complex resources of an application into a single implementation unit called a Chart.

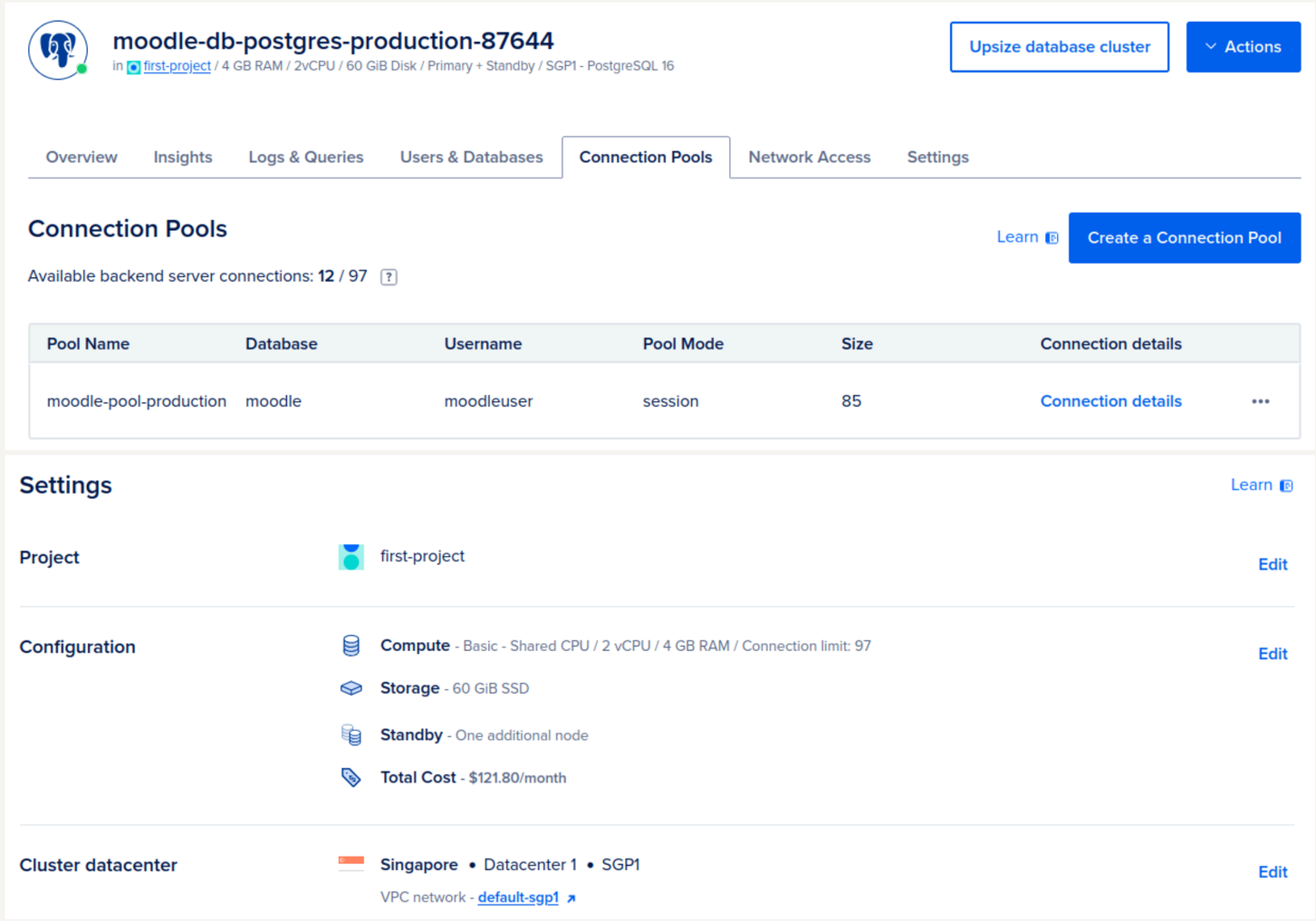
Criteria	Manual YAML	Helm
Resource Management	Decentralized	Centralized
Reusability	**Low** (Recreated for each environment)	**Very high** (Shared templates)
Deployment Operation	Run commands per file/folder	Single-command deployment
Version Control	Difficult (Manual tracking)	Built-in revision management & rollback
Suitability	**Simple** architecture	**Complex**

Database

- Master - Replica (Write - Read)
- **Session** mode instead of **transaction** mode
- Deploy **PgBouncer** for connection stability
- Managed Database (default):
  - Monitors and alerts admins via email when disk space is full.
  - **No** Auto-scaling storage

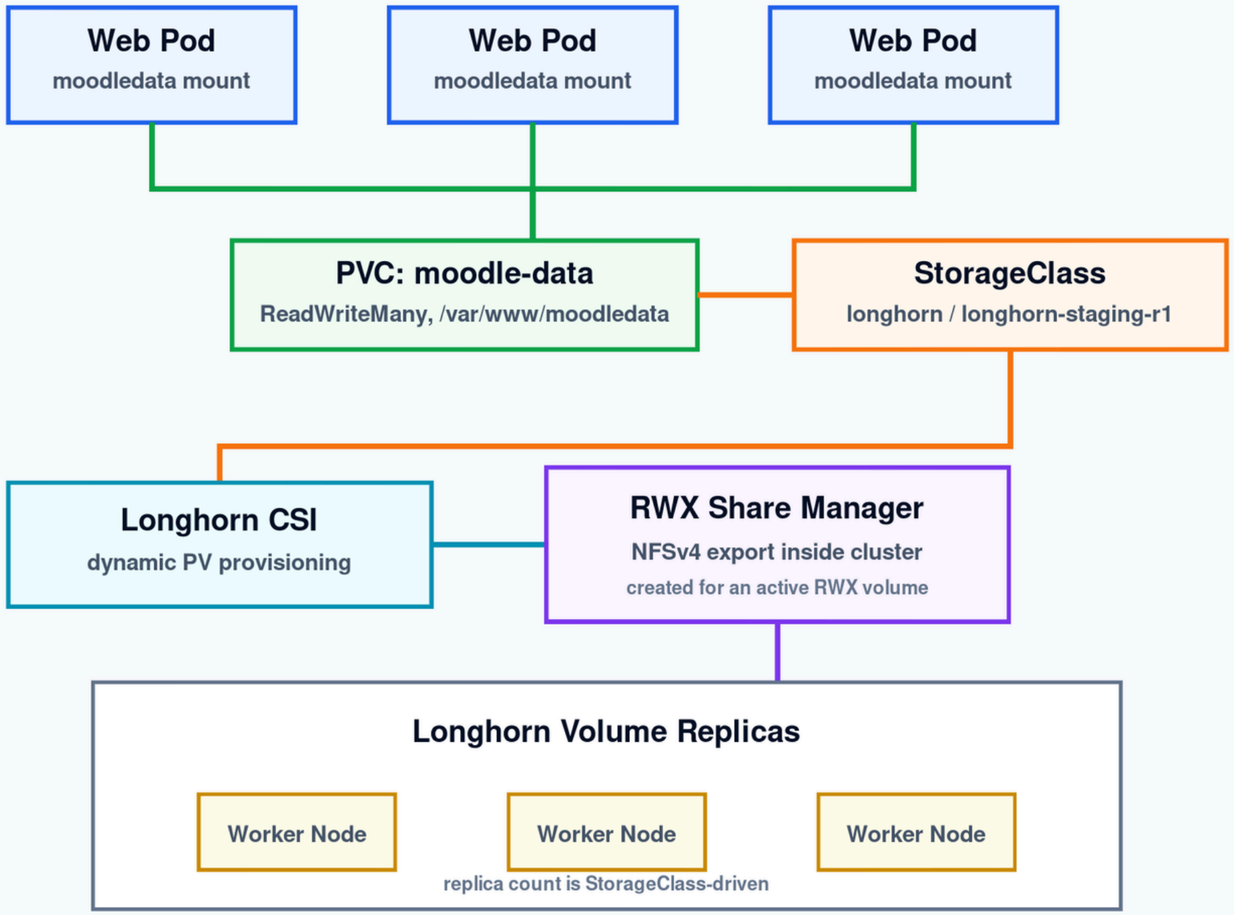
(**Relies entirely** on the cloud provider).

PostgreSQL



Config & Storage Resources

SHARED FILE STORAGE PATH



RESOURCE PLACEMENT

Longhorn RWX PVC
- Shared Moodle files and uploads
- Mounted at /var/www/moodledata

Pod-local runtime
- Moodle app copy on emptyDir
- /tmp/moodlelocalcache
- OPcache and APCu in memory

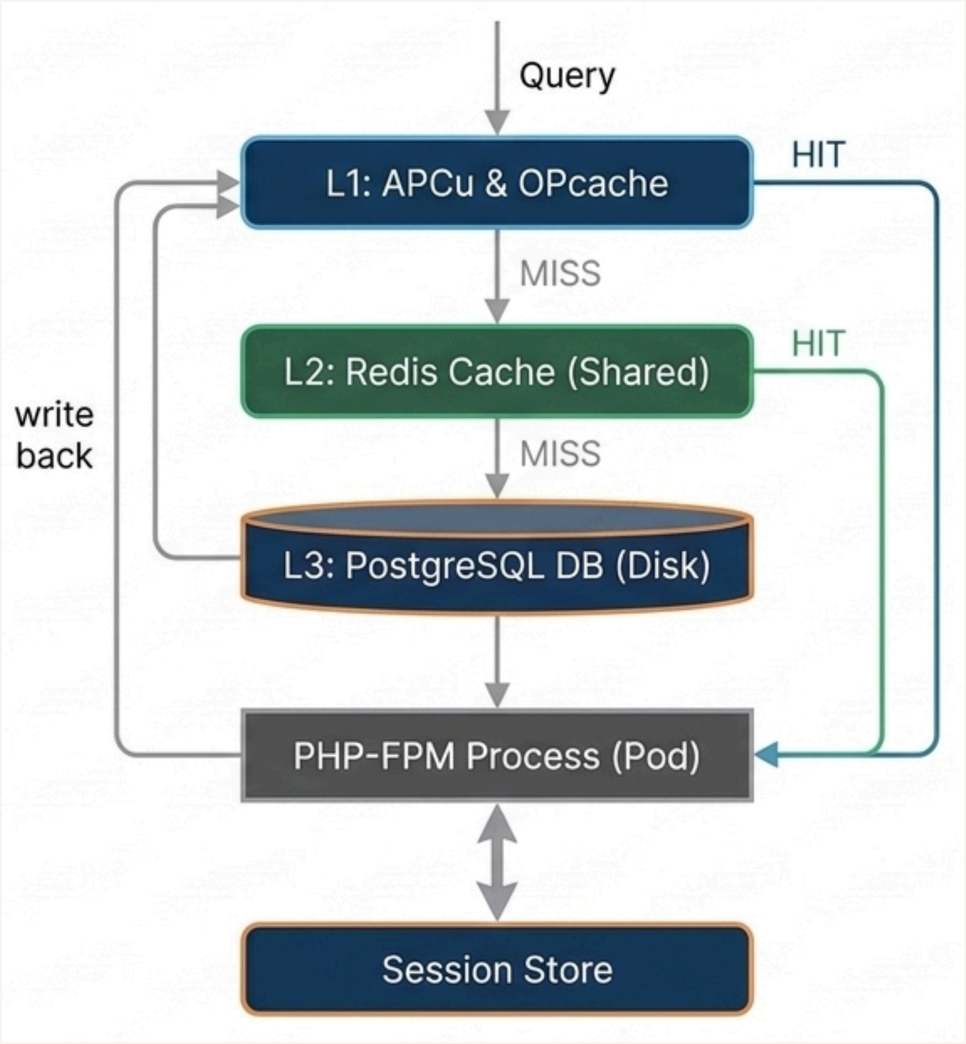
ConfigMap / Secret
- ConfigMap: NGINX and cache setup
- Secret: database password

Database is separate
- DigitalOcean Managed PostgreSQL

Longhorn stores shared files, not PostgreSQL data.

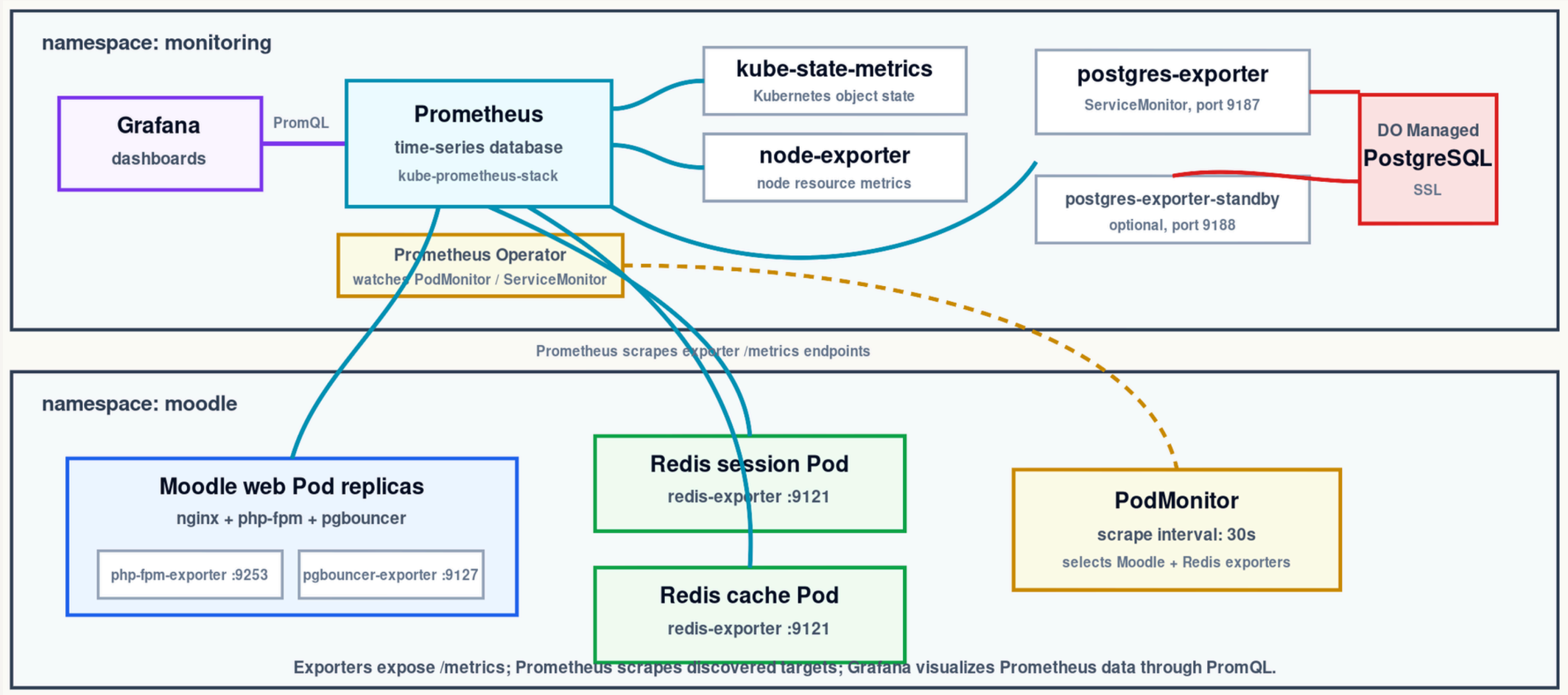
Caching

- **L1** Cache (Per-pod, RAM):
  - **APCu** cache
  - **OPcache** cache
- **L2** (Shared across nodes, RAM):
  - **Redis cache**
- **L3** Storage (disk):
  - **Database**
- **Session Store:** Stores user login sessions.



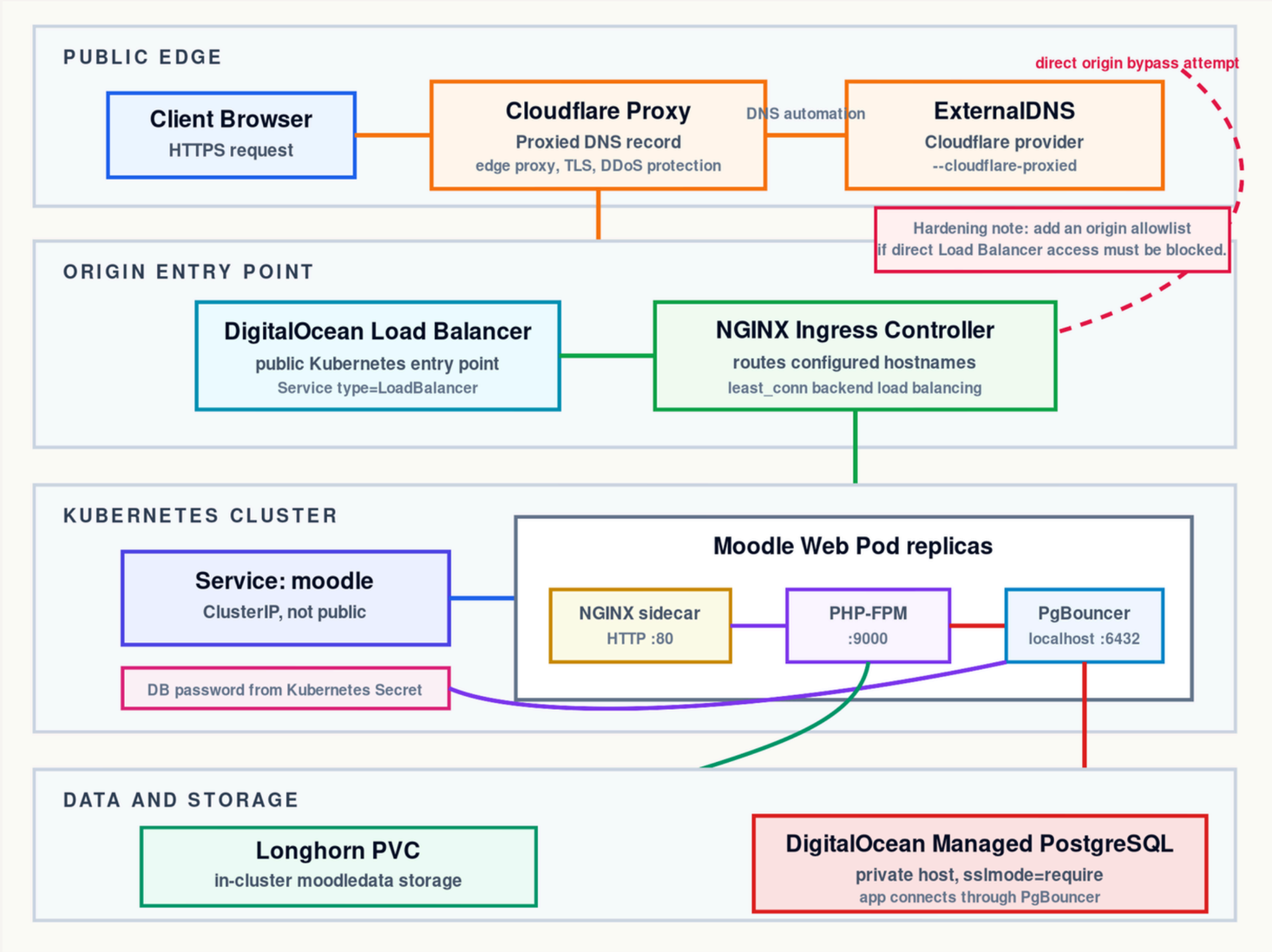
Observability & Metrics

- Separate Grafana into purpose-driven dashboards
- Exporters deployed per pod to collect metrics

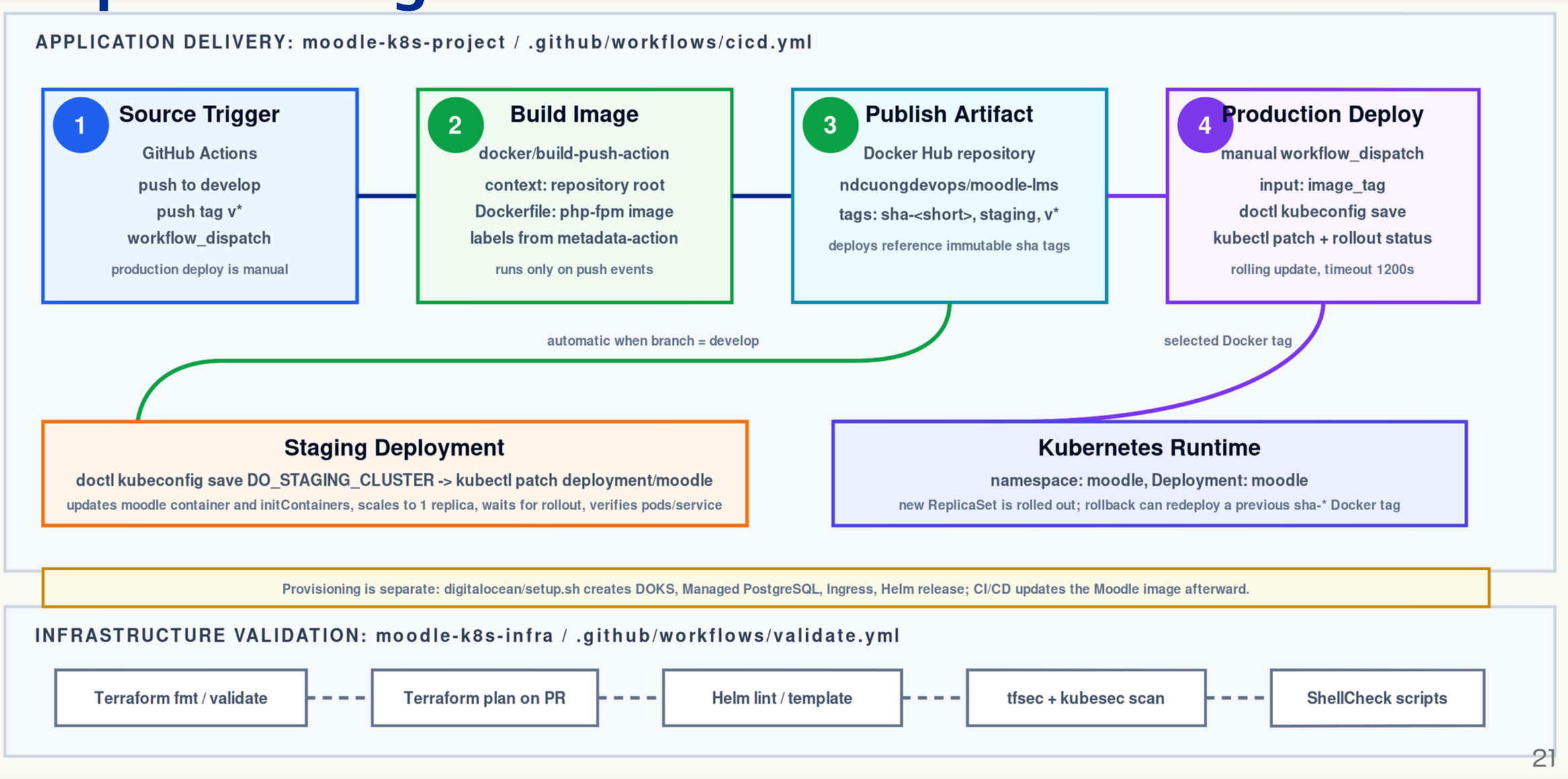


Security

- Infrastructure-layer attacks handled mostly by Cloudflare
- Moodle restricted to domain-only access, blocking IP
- Database restricted strictly to internal Kubernetes access



Implementing CI/CD



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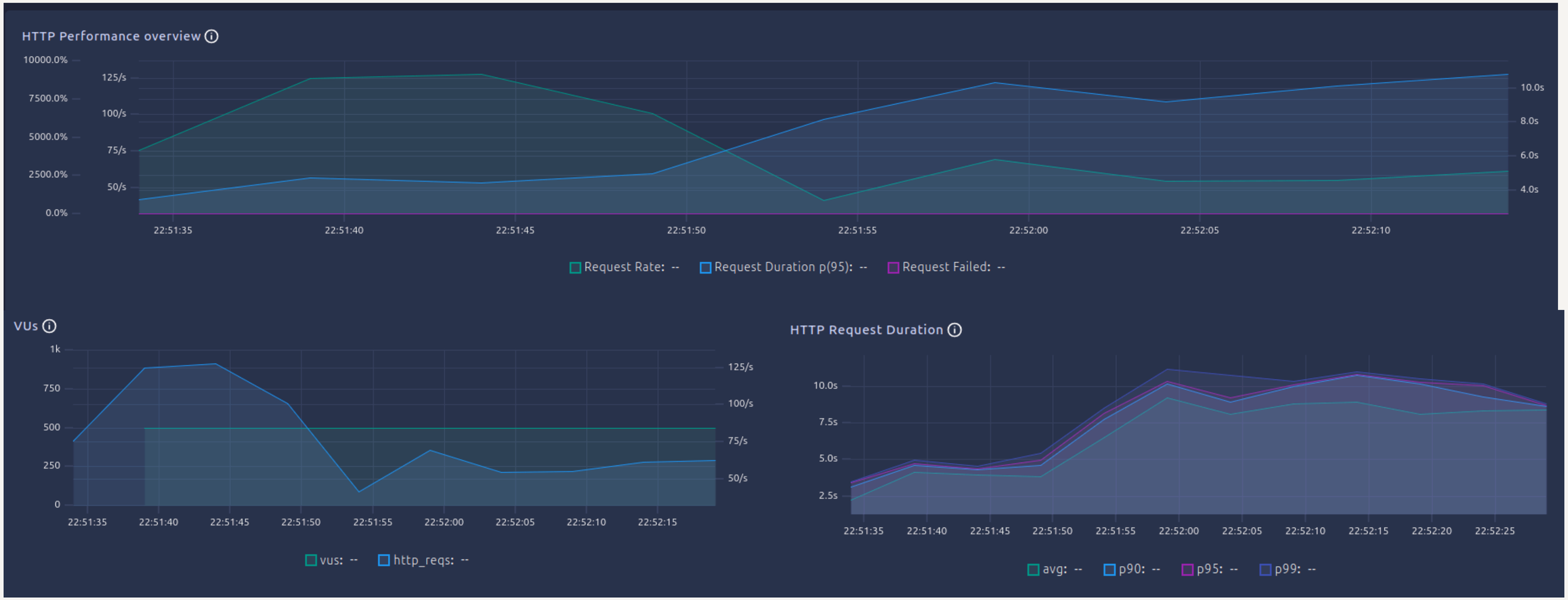
Results and future development

Load capacity

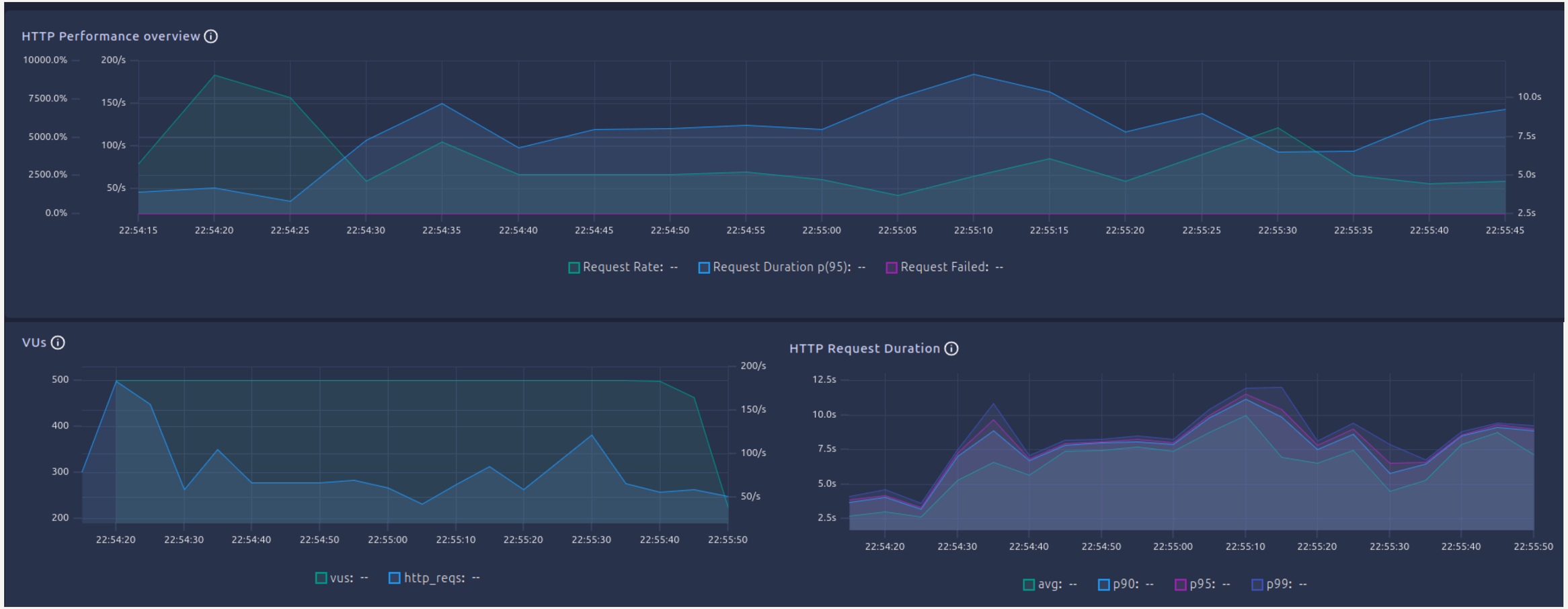
Scenario: 500 users access quiz, then simultaneously submit after 30s



First 30s: Users access quiz and pre-fill answers

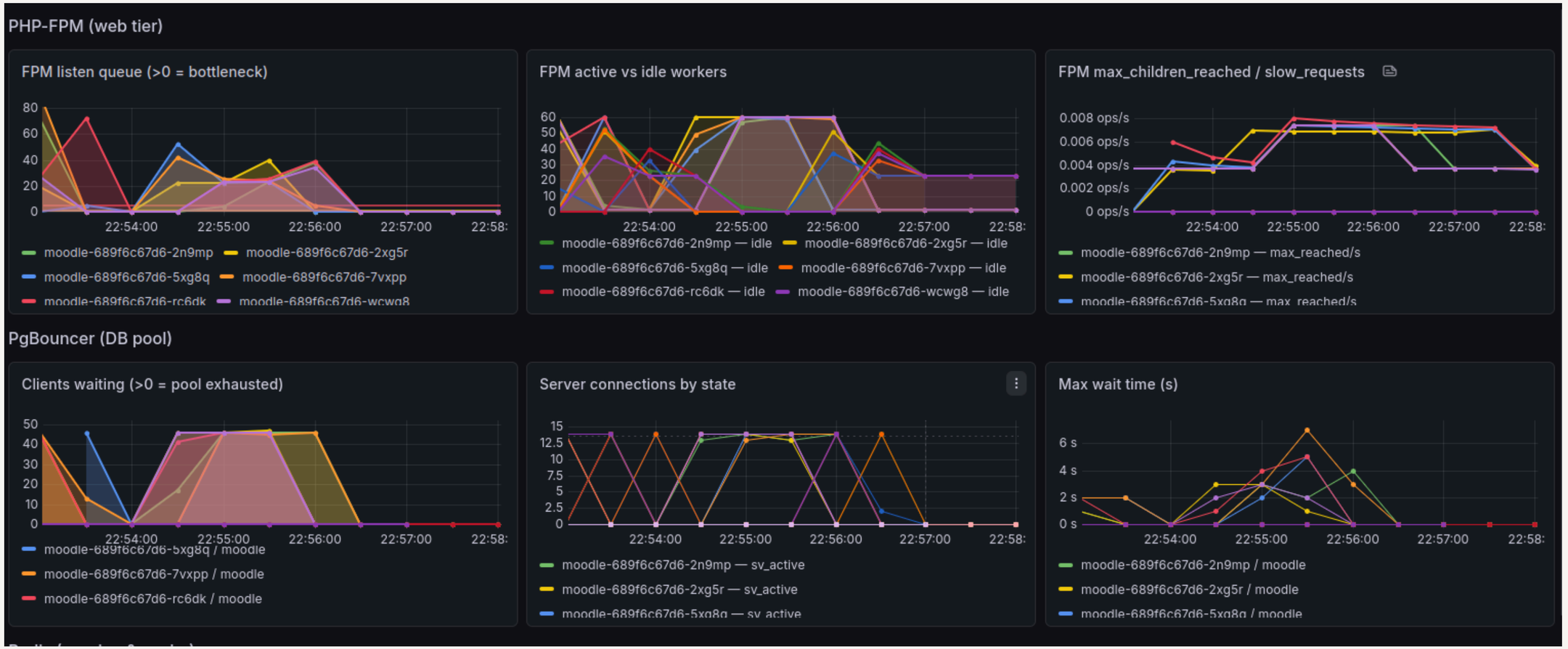
Load capacity

Scenario: 500 users access quiz, then simultaneously submit after 30s



After 30s: Users simultaneously submit quiz at the same time

Dashboard Grafana



Conclusion:

- Database bottleneck with high PgBouncer queue
- Blocked queries causing App-layer connection backlog

Constraints

- Only short-term testing conducted
- CI/CD limited to app and infrastructure
- High infrastructure cost for Moodle AI

Future Development Plans

- Enhance monitoring, alerting, and incident response
- Implement central logging for daily operations
- Deploy Cloudflare Waiting Room for downtime
- Optimize software layer over infrastructure layer

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DEMO

Thank You For Your Listening